

Bailey Jannuzzi

PhD Student, Mathematical and Computational Biology
Department of Applied Science
William & Mary

Education

William & Mary Applied Science Department PhD in Mathematical and Computational Biology Advisor: Geoffrey Zahn, PhD	Williamsburg, VA Expected: 2031
Utah Valley University Bachelor of Science in Bioinformatics Magna Cum Laude GPA: 3.81	Orem, UT Spring 2026

Research Experience

William & Mary — TIDAL Lab Graduate Research Assistant Research Advisor: Geoffrey Zahn, PhD Studying microbial community assembly, with a focus on applying graph neural networks to model how microbiomes coalesce. - Contributing to lab research on microbiome-based approaches to plant and ecosystem health, including metagenomics and machine learning methods.	Williamsburg, VA Fall 2026 - Present
Utah Valley University — Hjelman Lab Independent Student Researcher Research Advisor: Carl E. Hjelman, PhD Investigating the diet of forensically relevant blow flies (Diptera: Calliphoridae) to determine their effectiveness as wildlife monitors. - Constructing reproducible bioinformatic pipelines to identify environmental DNA from blow fly gut content samples.	Orem, UT Spring 2024 - 2026
Utah Valley University — Hjelman Lab Peer Research Collaborator Research Advisor: Carl E. Hjelman, PhD Assisting with gathering data on blow fly (Diptera: Calliphoridae) diversity and distribution in Utah. - Conducting field collections. - Preparing collected specimens through pinning and labeling to be identified.	Orem, UT Spring 2025 - 2026

Presentations

*Presenting author

Poster Presentations

American Society for Microbiology, Intermountain Branch Meeting <i>Evaluating Genomic Pipelines for Predicting Bacteriophage Host Range in Salmonella enteric</i> , *Rogers N, Jannuzzi B, Johnston M, Lopez A, Brooks L	2026
--	------

Entomological Society of America, Pacific Branch Meeting 2025
Oxford Nanopore Technology Sequencing and Identification of Fly Gut Content, *Jannuzzi B, Meeds A, Weidner LM, and Hjelman CE

Entomological Society of America, Annual Meeting 2025
From City to Canyon: A Survey of Blow Fly Biodiversity Across Utah Ecoregions, *Beck H, Jannuzzi B, Weidner LM, and Hjelman CE

Oral Presentations

North American Forensic Entomology Association, Annual Meeting | 1st Place Award 2026
From Gut to Database: Leveraging Blow Fly Dietary DNA for Forensic and Conservation Applications, *Jannuzzi B, Hornsby D, Meeds A, Weidner LM, and Hjelman CE

North American Forensic Entomology Association, Annual Meeting 2026
Characterizing Calliphoridae Diversity in the Mountain West: A Statewide Survey of Blow Fly Communities in Utah, *Beck H, Jannuzzi B, Weidner LM, Hjelman CE.

Entomological Society of America, Annual Meeting | 2nd Place Award 2025
Optimizing a Bioinformatic Pipeline for Profiling Blow Fly Diets, *Jannuzzi B, Meeds A, Weidner LM, and Hjelman CE

Entomological Society of America, Annual Meeting 2025
What flies eat and excrete: Insights from puke, poop, and gut contents of blow flies for wildlife and forensic applications, *Weidner LM, Jannuzzi B, Yeilee Perez Ortiz, Gabrielle Meza, Meeds A, and Hjelman CE

Awards and Funding

Funding (Totaling \$3,719)

Undergraduate Research, Scholarly, and Creative Activity Dissemination Grant 2025
 Amount Awarded: \$1,093

Scholarly Activities Committee Grant 2025
 Amount Awarded: \$1198

Undergraduate Research, Scholarly, and Creative Activity Grant 2025
 Amount Awarded: \$1428

Awards and Honors

NAFEA Annual Meeting | 1st Place Oral Presentation 2026

Outstanding Student Award for Biology — UVU, College of Science Spring 2026

ESA Annual Meeting | 2nd Place Oral Presentation 2025

Dean's List — Utah Valley University Spring 2023 - 2026

Experience

William & Mary Graduate Teaching Assistant Supercomputing (APSC 624/424) - Grading coursework and assignments - Assisting students with questions on course material and assignments	Williamsburg, VA Fall 2026
Utah Valley University Teaching Assistant General Biology (BIOL 1010) - Grading assignments and exams - Hosting weekly study and review sessions - Assisting with in-class activities - Teaching vignette-style lectures in class	Orem, UT Fall 2025
Utah Valley University Teaching Assistant Genetics (BIOL 3500) - Grading coursework and assignments - Supporting students in applying class concepts to solve practical problems	Orem, UT Spring 2025

Membership

W&M Biomath Journal Club/Seminar	Fall 2026 - Present
American Society for Microbiology, Intermountain Branch	Spring 2026 - Present
North American Forensic Entomology Association	Spring 2026 - Present
Entomological Society of America	Fall 2025 - 2026
UVU Evolution and Bioinformatics Club President	Spring 2025 - 2026
Entomological Society of America, Pacific Branch	Spring 2025 - 2026
UVU Evolution and Bioinformatics Club Secretary	Spring 2024 - 2025

Skills

Computational Skills R, Python, SQL, BASH Scripting/Linux, Bioinformatic Pipeline Development (Oxford Nanopore sequencing, eDNA metabarcoding)
Laboratory Skills DNA Extraction, PCR, gel electrophoresis, DNA purification, insect specimen preparation (pinning & labeling)